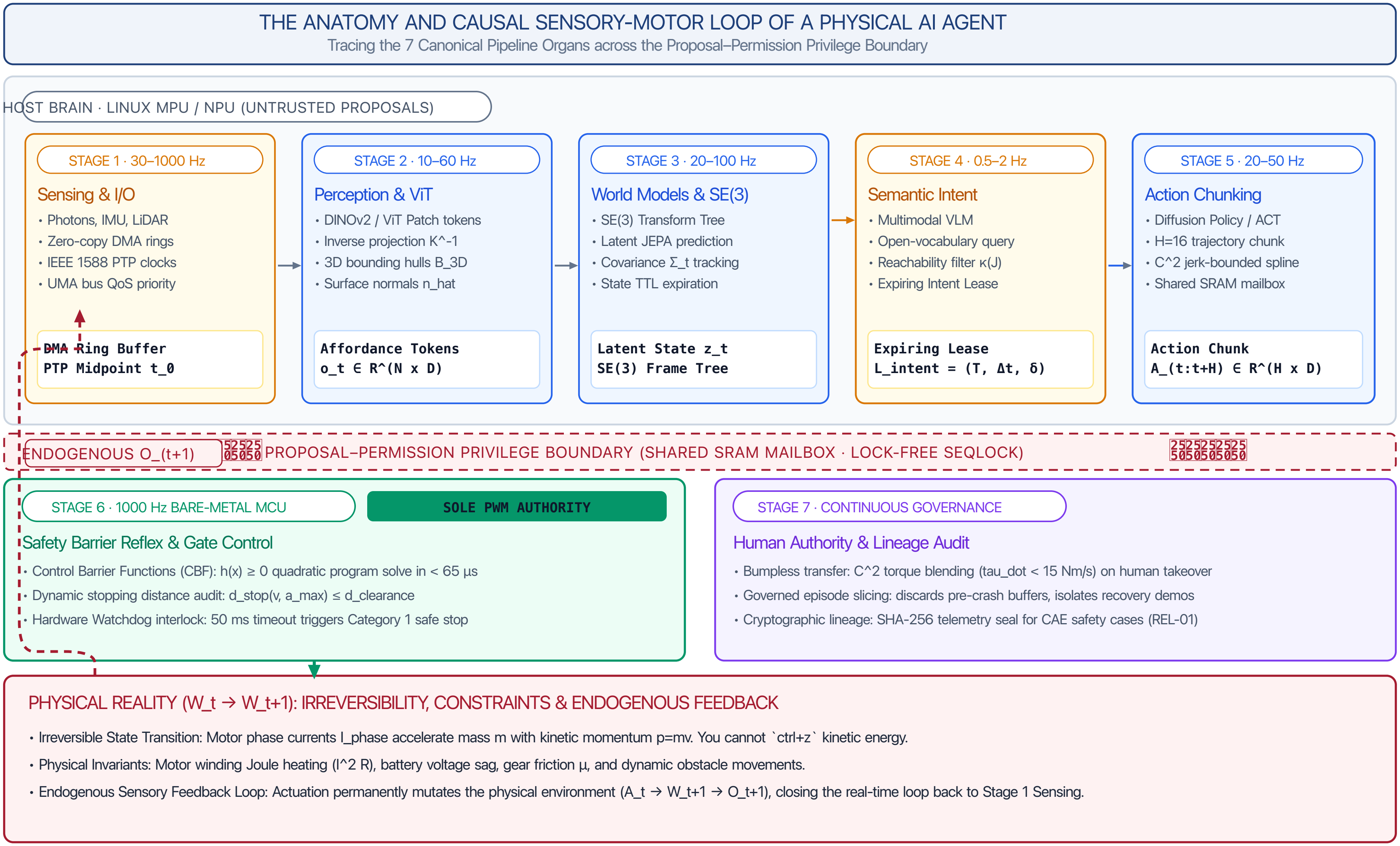
THE ANATOMY AND CAUSAL SENSORY-MOTOR LOOP OF A PHYSICAL AI AGENT
Tracing the 7 Canonical Pipeline Organs across the Proposal–Permission Privilege Boundary
HOST BRAIN · LINUX MPU / NPU (UNTRUSTED PROPOSALS)
STAGE 1 · 30–1000 Hz
Sensing & I/O
• Photons, IMU, LiDAR
• Zero-copy DMA rings
• IEEE 1588 PTP clocks
• UMA bus QoS priority
DMA Ring Buffer
PTP Midpoint t_0
STAGE 2 · 10–60 Hz
Perception & ViT
• DINOv2 / ViT Patch tokens
• Inverse projection K^-1
• 3D bounding hulls B_3D
• Surface normals n_hat
Affordance Tokens
o_t ∈ R^(N x D)
STAGE 3 · 20–100 Hz
World Models & SE(3)
• SE(3) Transform Tree
• Latent JEPA prediction
• Covariance Σ_t tracking
• State TTL expiration
Latent State z_t
SE(3) Frame Tree
STAGE 4 · 0.5–2 Hz
Semantic Intent
• Multimodal VLM
• Open-vocabulary query
• Reachability filter κ(J)
• Expiring Intent Lease
Expiring Lease
L_intent = (T, Δt, δ)
STAGE 5 · 20–50 Hz
Action Chunking
• Diffusion Policy / ACT
• H=16 trajectory chunk
• C^2 jerk-bounded spline
• Shared SRAM mailbox
Action Chunk
A_(t:t+H) ∈ R^(H x D)
ENDOGENOUS O_(t+1)
PROPOSAL–PERMISSION PRIVILEGE BOUNDARY (SHARED SRAM MAILBOX · LOCK-FREE SEQLOCK)
STAGE 6 · 1000 Hz BARE-METAL MCU
SOLE PWM AUTHORITY
Safety Barrier Reflex & Gate Control
• Control Barrier Functions (CBF): h(x) ≥ 0 quadratic program solve in < 65 µs
• Dynamic stopping distance audit: d_stop(v, a_max) ≤ d_clearance
• Hardware Watchdog interlock: 50 ms timeout triggers Category 1 safe stop
STAGE 7 · CONTINUOUS GOVERNANCE
Human Authority & Lineage Audit
• Bumpless transfer: C^2 torque blending (tau_dot < 15 Nm/s) on human takeover
• Governed episode slicing: discards pre-crash buffers, isolates recovery demos
• Cryptographic lineage: SHA-256 telemetry seal for CAE safety cases (REL-01)
PHYSICAL REALITY (W_t → W_t+1): IRREVERSIBILITY, CONSTRAINTS & ENDOGENOUS FEEDBACK
• Irreversible State Transition: Motor phase currents I_phase accelerate mass m with kinetic momentum p=mv. You cannot `ctrl+z` kinetic energy.
• Physical Invariants: Motor winding Joule heating (I^2 R), battery voltage sag, gear friction µ, and dynamic obstacle movements.
• Endogenous Sensory Feedback Loop: Actuation permanently mutates the physical environment (A_t → W_t+1 → O_t+1), closing the real-time loop back to Stage 1 Sensing.